Sinusoids

Copyright © Barry Van Veen 2016

AllSignalProcessing.com



1. Frequency is _____ proportional to fundamental period.

    a) directly

    b) inversely

2. Continuous-time sinusoids with distinct frequencies are _____.

    a) always distinct

    b) not necessarily distinct

3. Discrete-time sinusoids with distinct frequencies are _____.

    a) always distinct

    b) not necessarily distinct

4. Discrete-time frequency is measured in units of _____.

    a) Hz

    b) rads/sec

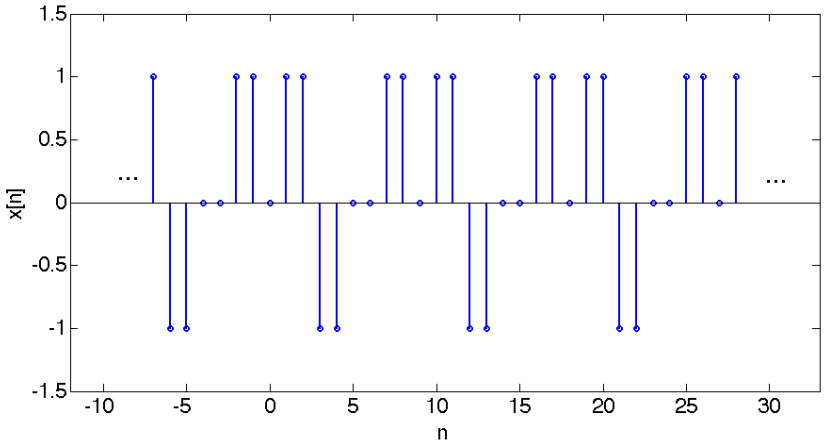
    c) rads

5. A complex sinusoid $x[n] = Ae^{j\pi n/8}$ has real part given by a _____.

    a) cosine

    b) sine

6. Find the fundamental frequency of the signal depicted below.



7. Identify which of the following discrete-time sinusoids are identical

   a) $x[n] = 5\cos(\frac{\pi}{4}n)$

   b) $x[n] = 5\cos(\frac{3\pi}{8}n)$

   c) $x[n] = 5\cos(\frac{19\pi}{8}n)$

   d) $x[n] = 5\cos(\frac{19\pi}{4}n)$

   e) $x[n] = 5\cos(\frac{9\pi}{8}n)$

   f) $x[n] = 5\cos(\frac{21\pi}{8}n)$

   Verify your answer using DTSinusoid.m.

8. A continuous-time sinusoid $x(t) = 4\cos(\Omega_o t + \pi/8)$ is sampled to produce a discrete-time sinusoid $x[n] = x(nT)$. Find the frequency ω_o of the discrete-time sinusoid for the following choices of Ω_o and T.

a) $\Omega_o = 4\pi$ rads/sec, $T = 0.1$ secs

b) $\Omega_o = 4\pi$ rads/sec, $T = 0.2$ secs

c) $\Omega_o = \pi$ rads/sec, $T = 0.1$ secs

d) $\Omega_o = 100\pi$ rads/sec, $T = 10^{-3}$ secs

9. The fundamental frequency of a trumpet note $x(t)$ is f_c Hz.

a) Find the fundamental frequency in units of rads/sec if $f_c = 880$ Hz.

b) You sample $x(t)$ at an interval $T = 10^{-4}$ sec to obtain $x[n]$. What is the fundamental frequency of $x[n]$ if $f_c = 880$ Hz.

c) You sample $x(t)$ at an interval $T = 10^{-4}$ sec to obtain $x[n]$. What is the fundamental frequency of $x[n]$ if $f_c = 1760$ Hz.

10. Write $x(t) = 4\cos(120\pi t + \pi/3)$ as a sum of two complex sinusoids.

11. Write $x[n] = je^{j\pi n/8} - je^{-j\pi n/8}$ as a real-valued sinusoid.

12. Find the fundamental period for each of the following signals.

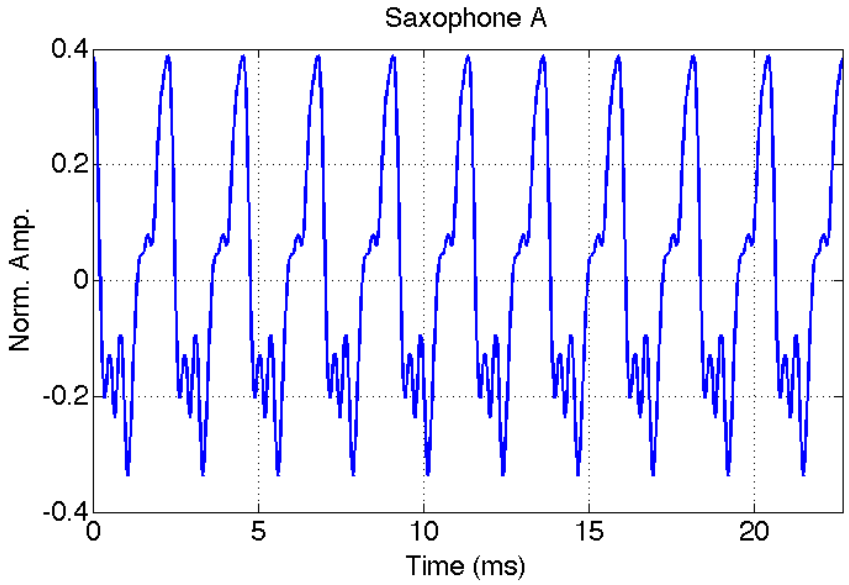
    a) $x(t) = \cos(\pi t)$

    b) $y(t) = \sin(2\pi t/3)$

    c) $z(t) = x(t) + y(t)$

Hint: if a signal $x(t)$ has fundamental period T_x, then it also is periodic with period mT_x where $m = 1, 2, 3, \ldots$.

13. The sound file SaxA.wav contains a brief recording of an alto saxophone playing the musical note A. Listen to the sound. A 22.65 ms section of this note is depicted below.



Use the graph to estimate the fundamental frequency of this note.

14. The m-file SaxAapprox.m will display the approximation of this note as a sum of eight harmonically related cosine signals. Signals are harmonically related if they are integer multiples of a common fundamental frequency. The kth harmonic of fundamental f_o has frequency kf_o. Listen to the approximations using 1, 2, 3, 7, and 8 sinusoidal components and compare them to the original. Use the data cursor to find the approximate amplitude of the following cosines in the approximation.

a) $k = 2$

b) $k = 5$